

Video Transcript

Video 4: Mapbox Overview (05:53)

Maps are a great example of bringing in resources onto our page and mixing them up with our UI. We can bring in the libraries, we can bring in the styles, and we can layout great resources in this way. In this case, we're going to bring in maps, and so we're going to start off by bringing in the library, and that is Mapbox.com's library as you can see there, and we're using this one because it allows you over 15,000 or 50,000 actually views per month for free. This is the library. Now, I'm going to go ahead and bring in the styles.

This brings in the styles for the mapping library, and we can view either one of these by navigating directly in the browser. So, in this case, we're looking at the mapping library, and you can see there how big it is, even though it's minimized. And I could do the exact same thing for the styles by accessing them directly. So, by doing those includes in the HTML document, we're loading all of those resources, and they will now be available for us to use within our page.

Next, I'm going to add some styles. I'm going to set some styles for the body because I will be laying out a map. I want my margins and my padding to be zero, to make sure I have no spacing around the map. Then, the styles for the map element are going to be the following. I'm going to set my position to be absolute, my top at zero, bottom at zero, and my width to be a 100%. Now within the body of the document, I'm going to create the element that will hold the map and I'm going to use a div for this, and the ID of the div is going to be map. When we write our code in our script block, we will reference that div. We're now going to use a function within the mapping library to create a new map. And so I'm going to use 'new' Mapboxgl.Map(), and this creates, this function creates a new map, and that function takes an object that has a certain number of properties. So, the first one is 'container'. And here is where we reference '.div' ID of map.

And so, I'm going to enter that here. Next, I'm going to set the style, and this is defined by the library and the styles makers which in this case is mapbox. So, I'm going to go ahead and paste that in here. Next, we're going to set the starting location, which is going to be the longitude and latitude. And this is one thing to watch, depending on which library you're using, you could get

latitude first or longitude first. And then, the last thing that we're going to set is the initial zoom on the page. So, in this case, we're going to set the initial zoom at 10. And we will change these in a moment once we load the map. But there's one more thing that we need. The additional thing that we need is that we need an access token.

And here, I will refer you to account.mapbox.com for you to get your own access token. I will not share mine because since these videos are viewed by many, they would stop working soon after the 50,000 limit per month is reached. So, I'm going to go ahead and paste my token here and now let's go ahead and load up the map. So, I'm going to go ahead and reload this page and as you can see there we are viewing the city of Boston. If I was to zoom in, let's say we use 15 here, I'm going to go ahead and reload the page, then you could see that we are zoomed in to Cambridge and we are at MIT's campus. I could change the styles as well. Let's say we go with a dark style instead of the default one that I'm using here, and let me go ahead and remove all of it and paste the new style. And now, I'm going to go ahead and reload the page. And you'll see there that now the styles are very different.

So, this is an example of basic mapping. If you use the Google Maps API, you would do something similar, as I mentioned as well, Bing would do it as well, and many others. You can see here some of the elements are bringing in the library, bringing in the styles, setting some of the styles around the element that you're going to set. We talked about margining and padding, and other things, and then creating the function or using the function that is going to create the map, giving it the element within the page, in this case, the div id map that where you're going to position your map. And then some of the initial settings as well, which are the level of zoom, the location and latitude and longitude, the style that you're going to use, and in this case the container. So, give it a try. This is something that is very useful in many different contexts, and as you can see, you can control this programmatically.